

Exploratory Experiments in Deep-Learning Methods

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<https://github.com/syeddanialg/pa0-atml>

Abstract

This report presents four exploratory experiments covering supervised, transformer-based, contrastive, and generative-latent deep learning paradigms. First, a ResNet-152 is studied on CIFAR-10 under frozen-head training, disabled residual connections, layer-wise feature separability, and three transfer-learning regimes (head-only, full-backbone, random initialization). Second, a pretrained Vision Transformer (ViT-B/16) is examined through attention-map interpretation, patch-masking robustness, and CLS-token versus mean-pooled classification. Third, CLIP (ViT-B/32) is evaluated for zero-shot classification on STL-10 under different prompt styles, and the image-text modality gap is analyzed and partially closed using orthogonal Procrustes alignment. Fourth, a convolutional Variational Autoencoder is trained on MNIST, and its latent space, reconstructions, and generations are compared against Doersch’s tutorial implementation. Across all experiments, results consistently confirm that architectural choices (residual paths, attention, contrastive objectives, KL regularization) and training strategy jointly determine both optimization stability and downstream generalization.

1 Introduction

Deep learning spans supervised, self-supervised/contrastive, and generative paradigms, each with distinct optimization behavior and failure modes. This report documents four self-contained experiments carried out to build practical intuition about these paradigms: (i) transfer learning and residual connections in ResNet-152, (ii) attention-based reasoning in Vision Transformers, (iii) contrastive vision-language alignment in CLIP, and (iv) latent variable generative modeling with a VAE. Each section states the setup, reports quantitative results observed from training logs, and discusses the underlying reasons for the observed behavior.

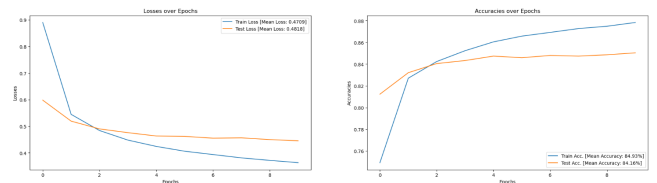
2 Task 1: ResNet-152 on CIFAR-10

2.1 Head-Only Baseline

A pretrained (ImageNet-V2) ResNet-152 was used with all convolutional layers frozen; only the final fully-connected layer (20,490 parameters out of 58.16M) was trained for 10 epochs with Adam ($lr = 10^{-3}$) on CIFAR-10. Training loss decreased monotonically while validation loss plateaued early, consistent with the classifier head quickly fitting a fixed feature extractor.

Epoch	Train Loss	Test Loss	Train Acc.	Test Acc.
1	0.891	0.599	74.92%	81.23%
5	0.424	0.464	86.05%	84.74%
10	0.363	0.446	87.84%	85.04%

Table 1: Head-only fine-tuning of ResNet-152 on CIFAR-10.



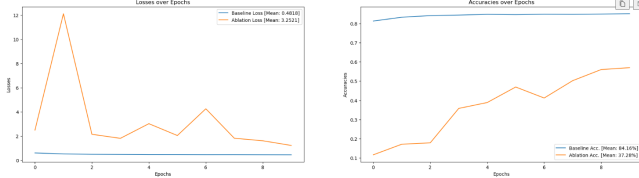
Training a 58M-parameter network from scratch on 50k images is impractical: the model overfits before it learns generalizable filters, and it is computationally wasteful. Since the head-only model reaches 85% accuracy with 0.04% of parameters trainable, the frozen early/mid layers already encode transferable low- and mid-level visual features (edges, textures, shapes).

2.2 Residual Connections

A second ResNet-152 was modified to bypass the additive skip path in the residual blocks of `layer1`, reverting them to plain stacked convolutions, and trained under the same protocol.

Variant	Test Acc.
With skip connections	85.04%
Without skip connections	56.99%

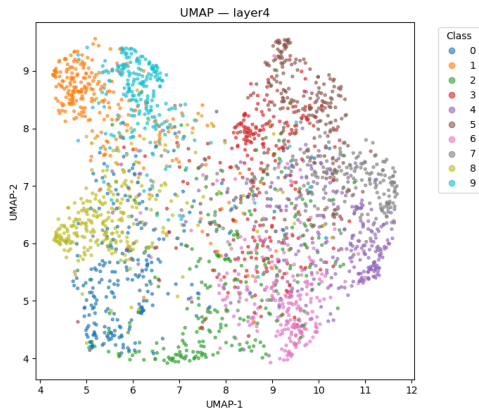
Table 2: Effect of disabling residual connections.



Removing the identity path eliminates the $+I$ term in the backward pass, $\partial y/\partial x = \partial F/\partial x$, so gradients must propagate purely through the convolutional Jacobian. Empirically this produced far noisier, slower convergence and a validation-loss spike at epoch 2 (12.12), confirming that residual paths stabilize gradient flow rather than merely accelerating training.

2.3 Layer-wise Feature Separability

Features from `layer1`–`layer4` of the fine-tuned ResNet-152 were pooled and visualized with UMAP over CIFAR-10 test samples.



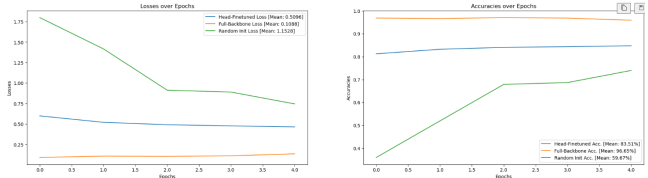
Class separability increases monotonically with depth: `layer1` embeddings form one intermixed blob with no visible per-class grouping, since early filters encode generic edges/textures shared across classes; separation becomes progressively clearer through `layer3` and `layer4`, where high-level, class-discriminative combinations of features emerge and distinct clusters are visible.

2.4 Transfer Learning Strategies

Three variants were compared on CIFAR-10: head-only fine-tuning, full-backbone fine-tuning, and training from random initialization, all under matched epoch counts.

Variant	Trainable	Val. Acc.	Time (5 ep.)
Head-only	20,490 (0.04%)	83.51%	~39 min
Full backbone	58.16M (100%)	96.65%	~1h43m
Random init	58.16M (100%)	59.67%	~2h

Table 3: Comparison of transfer-learning strategies (5 epochs each).



Trainable parameter count and accuracy do not scale together: full-backbone fine-tuning and random initialization update the identical 58.16M parameters, yet differ by 37 accuracy points, showing that additional trainable capacity only helps when optimization starts from a useful (pretrained) point in weight space. Head-only fine-tuning is the most compute-efficient option, while full-backbone fine-tuning gives the best accuracy at roughly $2.6\times$ the training cost of head-only.

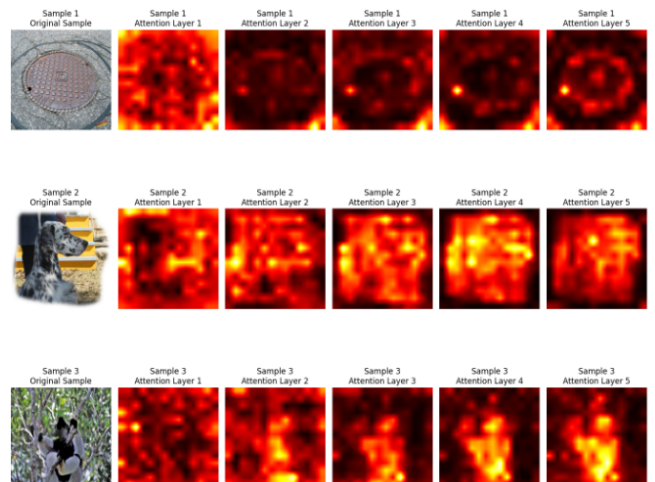
3 Task 2: Vision Transformers

3.1 Setup

A pretrained ViT-B/16 (google/vit-base-patch16-224, 12 layers, 12 heads, $d = 768$, patch size 16, $N = 196$ patches) was evaluated on a subset of ImageNet-1k validation images.

3.2 Attention Maps

Attention from the [CLS] token to all patches was averaged across heads and overlaid on input images.



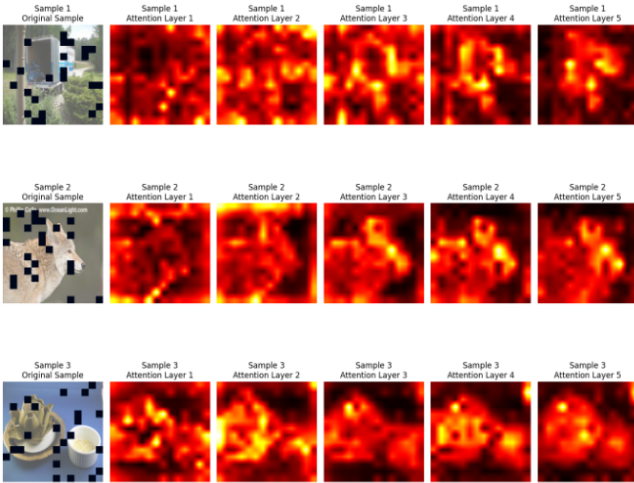
Early layers (1–4) attend broadly across the image; by layer 5 attention sharpens around the target object’s contour; layers 6–12 concentrate on a small number of fixed, semantically anchored spatial locations. This differs qualitatively from CNN saliency methods (CAM/Grad-CAM), which derive a single post-hoc localization map from gradients, whereas ViT attention is intrinsic to the forward pass and available per-head and per-layer.

3.3 Patch Masking

Patches were zeroed either randomly or in a structured central block, and classification accuracy on 128 samples was recorded (baseline: 100% on the 5-image sanity check; 128-sample batch used for masking).

Masking Strategy	Accuracy
No masking	100.00% [†]
Random patch masking	80.47%
Structured (center) masking	67.19%

Table 4: ViT accuracy under patch masking ([†]5-sample sanity batch).



ViT degrades far less under random masking than under structured center masking, because self-attention lets every surviving patch attend globally to every other surviving patch — unlike CNNs, whose local receptive fields lose spatial context irrecoverably. Structured masking is more damaging since it removes a spatially contiguous, often information-dense region (typically containing the object) rather than a scattered sample of patches.

3.4 CLS Token vs. Mean Pooling

Two classifier heads were compared on the full validation split (313 batches): one using the [CLS] token representation, the other using mean-pooled patch embeddings (with the classifier head fine-tuned for 5 epochs).

Pooling Method	Accuracy
CLS token	80.72%
Mean pooling (fine-tuned)	65.36%

Table 5: Pooling strategy comparison for downstream classification.

The CLS token outperforms mean pooling because it is explicitly optimized during pretraining to aggregate globally relevant information via self-attention, producing a task-tuned summary; naive averaging instead dilutes discriminative signal with background-patch information.

4 Task 3: CLIP

4.1 Zero-Shot Classification on STL-10

CLIP (ViT-B/32) was evaluated zero-shot on the STL-10 test set (10 classes) using three prompt styles.

Prompt Style	Accuracy
Plain class label	96.28%
“a photo of a {class}”	97.36%
Detailed/descriptive prompt	93.04%

Table 6: Zero-shot STL-10 accuracy under different prompt templates.

All three prompt styles remain within a narrow 4-point band, confirming that CLIP’s contrastive pretraining yields prompt-robust zero-shot generalization; naturalistic “photo of” phrasing performs best, plausibly matching the caption style most common in its web-scale training data, while overly verbose prompts slightly dilute the class signal.

4.2 Modality Gap

Image and text embeddings (both L2-normalized) were jointly projected with UMAP.

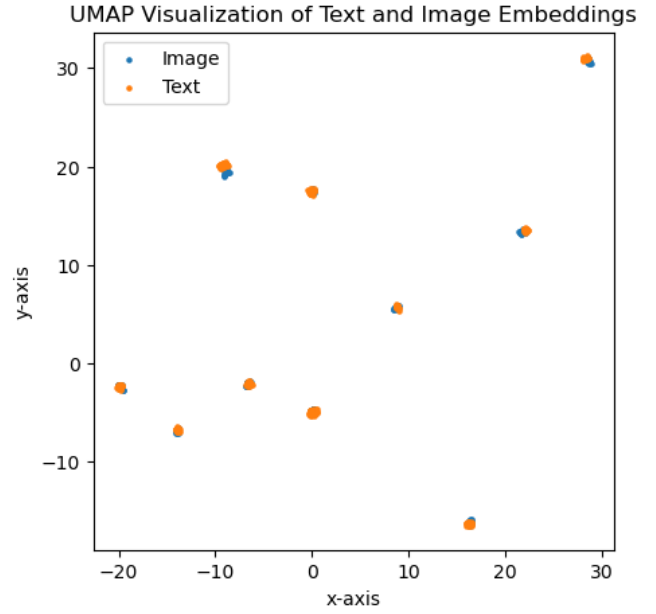
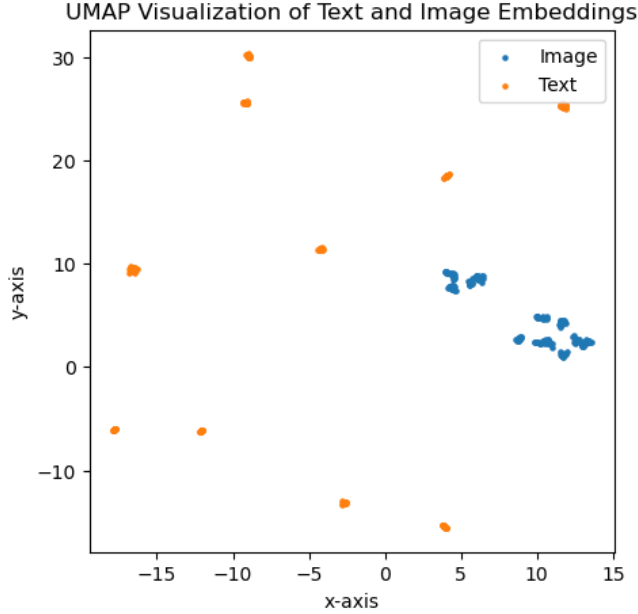


Image and text embeddings occupy visibly disjoint regions of the shared space — the “modality gap.” Normalization only places embeddings on a common unit hypersphere; it does not force the two modalities into a shared sub-region, since the gap originates from differing encoder architectures and training dynamics rather than from scale.

4.3 Bridging the Gap with Procrustes Alignment

An orthogonal Procrustes rotation R was fit to align image embeddings to the text embedding space.

Configuration	Accuracy
Baseline (unaligned)	96.28%
Procrustes-aligned	97.74%

Table 7: Zero-shot accuracy before and after modality alignment.

After alignment, the previously separated clusters overlap almost completely, showing that the gap is largely a linear rotational/positional offset rather than a deep semantic mismatch. However, classification accuracy barely changes, indicating that CLIP’s zero-shot performance depends on relative cosine ranking between an image and its candidate captions, not on the absolute geometric proximity of the two modality clusters.

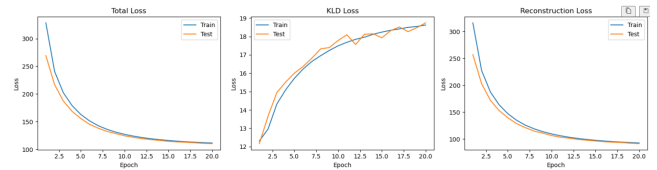
5 Task 4: Variational Autoencoder

5.1 Setup

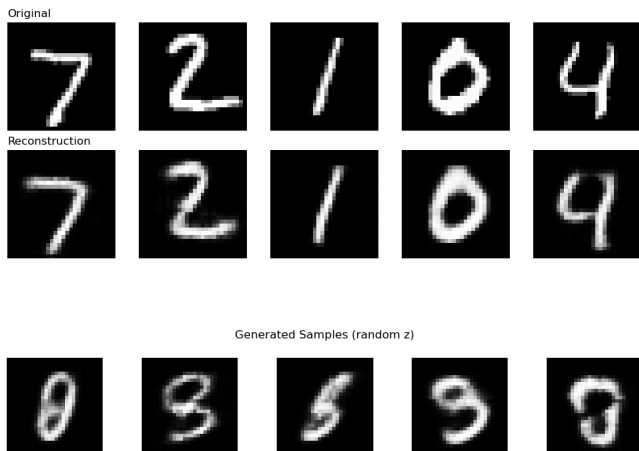
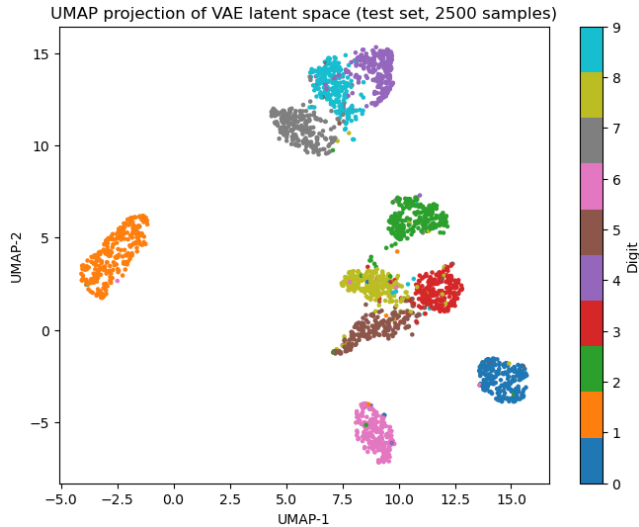
A convolutional VAE (3 encoder conv blocks with batch norm/ReLU/max-pool, mirrored transposed-convolution decoder, latent dimension 10) was trained on MNIST for 20 epochs (Adam, $lr = 10^{-4}$) with BCE reconstruction loss and standard KL regularization (no annealing).

Epoch	Total	KLD	Recon.	(Test) Total
1	328.44	12.31	316.13	269.37
5	162.97	15.72	147.25	155.67
10	127.17	17.49	109.68	124.49
20	111.50	18.62	92.88	110.40

Table 8: VAE training loss decomposition (BCE + KLD) over epochs.



5.2 Latent Space, Reconstructions, and Generation



The UMAP projection of the 10-dimensional latent space shows the ten digit classes forming well-separated clusters, despite no label supervision during training, indicating the encoder learns a semantically organized posterior. Reconstructions preserve digit identity and stroke shape but appear mildly blurred, a known consequence of the pixel-wise BCE objective. Random samples decoded from $z \sim \mathcal{N}(0, I)$ are mostly recognizable digits, with occasional ambiguous “in-between” shapes.

5.3 Comparison with Doersch’s Implementation

Doersch’s tutorial VAE is a fully-connected (MLP) network trained in Caffe with the same Bernoulli/BCE output assumption. The convolutional architecture used here is deeper and spatially aware, which should better capture local pixel-neighborhood structure than an

MLP of comparable size. Reconstruction quality and generation coherence were found to be comparable to Doersch’s reported results, including the presence of occasional blended, ambiguous digit samples — consistent with his observation that VAEs are relatively insensitive to latent dimensionality once it is neither too small nor too large.

6 Discussion

Across all four tasks, a common theme emerges: architectural mechanisms that provide either a direct optimization pathway (residual identity connections, self-attention over all patches, contrastive ranking losses, KL-regularized latent posteriors) consistently produce more stable training and better generalization than their ablated counterparts. Transfer learning experiments in Task 1 show that a good initialization matters more than raw trainable capacity. Task 2 shows that attention-based global context makes ViTs robust to missing local information in a way CNNs are not. Task 3 shows that contrastive alignment does not require the two modalities to occupy the same embedding region, only a consistent relative ranking. Task 4 shows that a simple KL-regularized latent variable model can organize class structure without supervision, at the cost of blurred reconstructions from the pixel-wise reconstruction loss.

7 Conclusion

These four experiments collectively illustrate core supervised, transformer, contrastive, and generative modeling principles: residual connections and pretrained initialization stabilize and accelerate CNN training; self-attention provides robustness and interpretability advantages over convolutional locality; contrastive pretraining yields prompt-robust, modality-gap-tolerant zero-shot classification; and KL-regularized latent variable models learn structured, class-separable representations from unlabeled data. Together, the results reinforce that both architectural design and training strategy jointly determine model behavior and downstream performance.